

What do anti-pain algorithms imply for computational functionalism?

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Abstract

We introduce a thought experiment on anti-pain algorithms that helps clarify competing theories of computational functionalism about consciousness. Under conventional interpretations of computation, it is natural to allow the existence of such algorithms: inverse functions that return a system exactly to the state it was in prior to some minimal pain algorithm.

The challenge lies in asking whether pain is felt upon the conclusion of the minimal pain algorithm but prior to the conclusion of the anti-pain algorithm. Both yes and no replies have intuitions in their favour but lead to options that constrain the nature of computationalist theories, motivating the value in the thought experiment for refining existing theories. If pain is felt before the anti-pain algorithm concludes, theorists are pressured either to permit micro-functions such that computational functionalism approximates phenomenal realism or to restrict consciousness only to systems that have system-wide visibility mechanisms. On the other hand, if pain is not felt, there are pressures to permit free-floating qualia, with only a tenuous link to the spatiotemporal physicality of its supervenience base.

Objections are discussed, including rejecting the possibility of reversible computation, whether in general or specifically for consciousness algorithms, and introducing notions of causality that move beyond a classical Turing Machine framework.

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1. Introduction

Is pain reversible? Not in the sense of nullifying the effects of pain with anaesthetic or dulling the memory of pain with time - but in the sense of an inverse function. All processes involved in the pain are exactly reversed, returning the system to its original state.

Under certain perspectives on consciousness and computation, such anti-pain algorithms are guaranteed to be possible in principle. Assuming those perspectives, our paper asks ‘whether pain is felt’ prior to a particular type of anti-pain algorithm being completed.

In conversation with the authors, some individuals have found it obvious that the original pain should be experienced in full during this period, while others found it equally obvious that pain would not be felt in certain circumstances, such as where the anti-pain algorithm completes quickly or begins instantly upon the onset of pain. It turns out there are natural intuitions pointing in both directions but that either option leads to curious consequences. In turn, owning those consequences brings precision to the types of consciousness theories possible under these perspectives. Because there are at least two mutually exclusive avenues to pursue, the anti-pain algorithm is a helpful forcing function for theorists to specify what they think happens and reflect on its implications for their theory.

This thought experiment is relevant to a diverse range of theories operating under computational functionalism (CF), the metaphysical assumption that implementing computations of a certain kind is necessary and sufficient for consciousness. Butlin et al. (2023) describe this assumption as a mainstream philosophical position, identifying aligned interpretations of major consciousness theories such as Global Workspace Theory, Recurrent Processing Theory, Predictive Processing frameworks, and others. We will follow this literature in assuming an emergent rather than a panpsychist view of whatever algorithms suffice for consciousness under CF. In other words, rather than a minimal act of information processing always being accompanied by experience (e.g. an individual bit flip), some degree of algorithmic sophistication is required before its function is accompanied by experience. We further restrict our scope to the physicalist interpretations of CF theories and to algorithms that can, in principle, be implemented on classical Turing machines (Hopcroft & Ullman, 1979), noting that it is possible to define computation in a broader manner (Shagrir, 1997).

CF is also indirectly supported by expert surveys and is the dominant assumption under which the possibility of artificial consciousness is currently being explored (Birch, 2025; Long et al., 2024). For instance, functionalism was the most popular account of consciousness in a 2020 survey, albeit still a minority position (Bourget & Chalmers, 2023). Meanwhile, 25% of AI researchers stated that some AI systems would be conscious by 2034 (Dreksler et al., 2025). The median estimate among 67 experts in another recent survey identified a 20% chance of digital minds having been created by 2030 (Caviola & Saad, 2025).

These theories comment on whether a given type of system might contain conscious experience. However, simply identifying plausible consciousness is of little value on its own - and arguably neither tractable nor necessary for questions of greater importance, such as what type of experiences are possible for that conscious entity (McClelland, 2025). If anti-pain algorithms exist and if the consensus position among relevant theories is that such algorithms prevent pain from having been felt, they represent a high priority research target for AI welfare programmes (Long et al., 2024).

The anti-pain thought experiment joins a rich heritage of thought experiments used for constructively exploring CF in three ways: arguing in favour of it, arguing against it, and seeking to refine it.

Arguments *in favour* of CF include multiple realizability (Bickle, 2020) and the dancing/fading qualia thought experiments (Chalmers, 1995) which argue for functionalism in general. The latter have been extended to CF (e.g. Hinton, 2025; Butlin et al., 2023 footnote 4), grounded in intuitions that emulating biological neurons with functionally identical alternatives with different causal mechanisms but the same computational input/output mappings at the neuron-level should not change the fact of consciousness in a system (although this extension can be challenged, e.g. Percy, 2026).

Thought experiments *against* functionalism (and therefore against CF) include the Chinese Room Argument (Searle, 1990) and the China Brain argument (Block, 1980). The former questions whether syntactic structures can ever generate semantic meaning or true understanding. The latter asks where

consciousness would reside if the relevant functions were emulated by large numbers of people signalling to each other via walkie-talkies.

Other thought experiments seek to *refine* CF, helping understand what versions of the position are plausible. Naturally, if a particular reader considers none of the available positions as plausible, they should reduce their confidence in CF as a valid assumption, but in general these arguments create space for at least one position that maintains the spirit of CF. Powerful examples include accepting the possibility of fractional qualia (Bostrom, 2006) or suggesting that counterfactual definitions of computation may not be appropriate given the Olympia thought experiment, which argues that counterfactual definitions would allow the presence of inert machinery to alter the fact of consciousness (Maudlin, 1989).

Our paper falls into this third category. If pain is felt before the anti-pain algorithm concludes, as we suggest is more likely, theorists are pressured either to permit micro-functions such that CF approximates phenomenal realism (with pain experienced as an absolute phenomenon regardless of any functional relevance) or to restrict consciousness only to systems that have system-wide visibility mechanisms to forecast likely functional relevance. On the other hand, if pain is argued not to be felt, theories risk permitting free-floating qualia, with only a tenuous link to the spatiotemporal physicality of its supervenience base. Multiple responses are viable, leading to different directions for future theory development.

Section 2 presents the anti-pain thought experiment, discussing the challenges faced by two possible responses to it (that pain is either felt or not felt prior to the anti-pain algorithm). Section 3 discusses possible escape routes and objections by rejecting the possibility of reversible computation, whether in general or specifically for consciousness algorithms, and options for rejecting the notions of causality applied in the thought experiment scenario. In the conclusion, we provide a table summarising possible responses to the thought experiment and touch briefly on other questions raised by it, such as implications for non-CF theories of consciousness, possible extensions to the idea, and what anti-pain qualia might feel like.

2. The thought experiment

This thought experiment is grounded in the work of Bennett (1973). Most general purpose computation is logically irreversible; consider how we cannot reconstruct a unique set of two inputs if all we know is that an AND gate returned 0. However, Bennett demonstrated that any Turing Machine equivalent calculation can be made logically reversible at every step, simply by saving the intermediate results at every step and, after finishing, “uncomputing” each step until the original state is restored. Bennett’s motivation was to identify the plausibility of more energy efficient thermodynamically reversible computers. In this paper, we use his result to motivate the existence of anti-pain algorithms in certain circumstances.

2.1. The anti-pain algorithm thought experiment setting

Assuming emergent computational functionalism (CF) about consciousness, there will be some minimal algorithm which generates a particular minimal pain experience for a given conscious entity (also defined by an algorithm) when run on a digital computer architecture from one point in time (t_0) to a later point in time (t_1). There might be a family of such minimal algorithms, but let us denote one such minimal algorithm as our target ‘pain algorithm’.

Given the nature of Turing Machine equivalent architectures, there will be at least one minimal inverse of this ‘pain algorithm’ (Bennett, 1973) that would undo the effect of the first algorithm. Allowing this minimal inverse algorithm to run from time t_1 to a later time (t_2), it returns the system to exactly its internal state prior to the first algorithm beginning (at time t_0), including options to dissolve any memory retained in the system or actions/action-planning initiated as a result of the first algorithm. Let this inverse algorithm with such options activated be called the ‘anti-pain algorithm’.

Assume that the t_1 - t_2 anti-pain algorithm is too quick for any actions initiated following the pain algorithm’s completion at t_1 to have achieved anything external to the system that is relevant to the system’s ordinary function. In other words, the external environment is unchanged as a result of the t_0 - t_2 algorithms beyond the passage of time and any thermodynamic noise dissipation required for the system to operate.

2.2. The necessary existence of ‘minimal’ algorithms

The thought experiment invokes the notion of a ‘minimal’ algorithm. This invocation merits defining and defending. A ‘minimal’ algorithm with respect to a given outcome is one where removing any part of that algorithm would prevent the outcome from occurring, i.e. the outcome cannot ‘partly’ occur partway through the algorithm. In this case, there can be no partial-pain felt or partial-anti-pain successfully applied before the minimal algorithms have completed in full. There might be more than one minimal algorithm for a given outcome; we only require that at least one exists.

The existence of ‘minimal’ multi-step algorithms is a direct consequence of adopting emergent CF and Turing Machine interpretations of algorithms, e.g. viable as finite calculations in discrete time. If partial-qualia are permitted for *arbitrary* partial completions of a minimal algorithm, then we must allow partial-qualia for individual computational steps (e.g. a single bit flip or single logic gate operation), which contradicts the distinction between emergent and panpsychist CF.

If we do not allow partiality down to individual computational steps, then we simply take one of those partial multi-step completions to be our minimal multi-step algorithm. Any partial amount of pain qualia still counts as qualia as far as this thought experiment is concerned and therefore we can ask about its corresponding inverse algorithm. Moreover, under the usual Turing interpretation of algorithms, the partial completion of an algorithm is itself an algorithm and would unlock any partial-qualia that correspond to it, noting that a midway completion step cannot be made dependent on the completion of future steps that have not happened yet, else we have simply reconstructed a full multi-step algorithm.

If we do allow partiality down to individual computational steps (such as a single NAND gate operation), we must acknowledge that such simple steps could easily be a constituent in all (or almost all) algorithms that generate the diverse forms of qualia. As a result, we have to allow that partial-qualia of all possible qualia variants supervene on each individual computational step. This implication goes beyond even the panpsychist variants of CF where each computational operation, however minimal, contains *some* minimal or basic conscious experience.

The implication of rejecting cleanly defined minimal algorithms under emergent CF is that the more basic the computational step, the more types of conscious experience it must contain! At the limit, each most basic step (e.g., a bit flip) contains, on its own, a flavour of every possible experience: a panpanpsychism, if you will, because every possible algorithm must contain at least one bit flip. While poetic, this conclusion seems unpalatable within canonical accounts of CF, which require

experiences to supervene on computations that have functional relevance for the system as a whole. As computations become simpler and have less functional import, CF does not expect experiential complexity to increase.

This implication also contradicts intuitions around information carrying capacity: how could a single bit flip contain information about every possible qualia? Presumably specifying each distinct qualia type requires some level of information under CF. The notion of partial-qualia might be conceivable, as might ontologically vague borderline conscious experiences (Schwitzgebel, 2023), but they do not fit within a single bit of information - at least not under a mainstream view of CF that invokes a Turing Machine equivalence framework with clean definitions of information and causal transitions. There might be niche CF positions which entirely decouple information volume from experiential complexity or which draw on quantum mechanical principles of ontological uncertainty, but such positions take us a long way from the Turing Machines or modern digital computers that motivate most contemporary CF theorists.

‘Minimal’ algorithms are also consistent with the idea of experience supervening on dynamic processes, provided those processes are fully captured in computable algorithms. For instance, under some theories, pain may not exist in single time-slices corresponding to the discrete point at which an algorithm concludes. Instead, pain only exists over extended periods of time, during which an algorithm is running dynamically, potentially with some core repeating component. In these cases, the ‘minimal’ algorithm would correspond to the dynamic algorithm having been sustained for the minimum period of time after which the system consciously experiences pain qualia. This might correspond to the minimum integration window after which the system experiences a detectable change in valence, analogous to the ‘just noticeable difference’ in perceptual states (e.g. Green, 2023). For humans this integration window may be on the order of hundreds of milliseconds, although it may differ substantially across architectures. Even with a vague onset boundary, i.e. no single time-step when we transition from ‘no pain qualia’ to ‘pain qualia’, the minimal algorithm can be taken to refer to the first time step after the vague boundary, beyond which pain is clearly being felt. Once this minimal dynamical process has completed, it is reasonable to ask if pain has been experienced even if the state were reversed thereafter.

As a result of these considerations, we conclude that ‘minimal’ multi-step algorithms for qualia are necessary components of emergent CF.

2.3. Was pain felt at t_1 ?

There are several probing questions that can be asked about the scenario in 2.1; we will focus on whether pain was felt at t_1 . We will argue that both ‘yes’ and ‘no’ answers have intuitions in their favour, but that each results in a different potential restriction or consequence for CF theories, such that this thought experiment can serve as a useful prompt for theorists to precisify their positions.

As an aside, we note that other options in multi-valued logic do not provide a simple exit. Partial-qualia and borderline-qualia are already excluded in this case via the ‘minimal’ algorithm rationale. Considering three-valued logic, answering the question with ‘unknown’ or ‘undefined’ might be reasonable from an observer’s perspective, but within the system there should be some fact of the matter about whether pain is felt - unless CF is redefined to treat minimal consciousness as a relational construct that exists only between multiple conscious entities rather than single entities.

Considering four-valued logic, answering the question with “both pain and not-pain” are felt (or “neither pain nor not-pain”) requires us to accept that computational systems are capable of qualia that

humans struggle to analogise. Indeed, the presence of such dual statements is considered sufficiently baffling that they are used in Buddhist koans to help meditators attain enlightenment and leave behind the humdrum world of ordinary logic. Asking (and expecting) the meditators' forgiveness, we will restrict this question to binary logic, noting that any existence of such dual qualia would prompt challenging questions for ethical theories that draw on feelings of pain or the inherent nature of any other feelings.

2.4. Why might pain be felt at t1?

From the perspective in which phenomenology is real in virtue of itself, then the answer is trivially 'yes': pain is felt at t1 for some duration specified by the algorithm - this follows by definition from the successful completion of a minimal pain algorithm. The anti-pain algorithm has not yet run, so cannot yet have had any effect on the system. Indeed at t=1 we cannot be certain the anti-pain algorithm will ever run or ever complete successfully, in which case the pain algorithm should have its usual phenomenal effect.

However, at face value, functionalism defines phenomenology in virtue of the functional role it plays in a system. The 'pain' at t1 achieves nothing functionally relevant for the system, so should not convey any experience. If nothing downstream differs, in what sense was there pain rather than a short-lived 'pain-shaped flurry of bits' that felt like nothing?

To sustain a 'yes' answer without abandoning the consequentialist aspect of functionalism, it seems we have to allow for the functional grounding of pain to operate either on very 'thin' functions or on very 'generous' counterfactual interpretations of the functions that begin to take place after t1 (i.e. the consequences of pain), before being undone by the anti-pain algorithm. Either approach prompts a useful clarification of a given CF position. Meanwhile, it seems unattractive to abandon the consequentialist requirement that conscious states are conscious in virtue of fulfilling a particular system-relevant functional role. Such a move shifts the theory towards a form of 'phenomenal realism', in which pain exists as a phenomenal object in its own right regardless of its role within the causal organisation of the system. This would represent a significant departure from the original CF motivations, which sought to explain consciousness in terms of the functional organisation of cognitive systems (although certain contemporary theories may be more open to such a move).

'Thin' functions

On the first option of 'thin' functions, during t1-t2 before the anti-pain algorithm is complete, we allow the pain algorithm's completion to initiate and progress other actions in the system (even if they are later reversed by the anti-pain algorithm). While these actions achieve no *complete* function of system relevance, they can still be interpreted as micro-functions and could convey micro-qualia. At the limit, however, we must allow the possibility of a rapid anti-pain algorithm - perhaps it can be parallelised with the pain algorithm such that the anti-pain algorithm was mostly pre-run in advance, with only a final operation or two required to finalise it after t1 (e.g. a single switch where all pain-affected memory units in the current state are converted instantly to what they would have been without the pain algorithm). As a result, we must allow very small micro-functions to qualify as 'thin' functions in this solution, perhaps as small as a single logic gate operation divorced from all consequences. This option seems to approximate a situation of phenomenal realism, in which the 'functional' aspects of CF have almost no system relevance, but still convey qualia.

Theorists could argue that micro-qualia do not matter and therefore do not challenge CF. For instance, they might point to examples such as a single miscoloured pixel in an image or a single millisecond of

sound; information that our conscious brains do not register even though it enters our early stages of information processing. However, by definition we require qualia to be felt (else they aren't qualia), so the examples are inappropriate. Even if instantly dismissed, we need sufficient pixels or milliseconds to trigger a tiny experience. And if feelings can be real without functional consequences, theorists have already accepted phenomenal realism. Moreover, we can construct a situation in which multiple micro-qualia happen immediately after each other. Imagine a series of separate sources that trigger pain algorithms, each with distinct plausible consequences that just get started, i.e. micro-functions get started (leading to brief duration micro-qualia) prior to each being undone by its own anti-pain algorithm. Having separate sources of relevance for the system, the consecutive micro-functions do not add up to a truly relevant system-wide consequence, but their micro-qualia are now plausibly part of a temporal sequence leading to pain experienced over non-trivial time.

'Generous' counterfactuals

On the second option of 'generous' counterfactuals, we could claim that micro-functions do not permit phenomenal experiences in their own right, but rather in virtue of the future consequences they enable to be possible.¹

Counterfactually, these micro-functions do lead to system-relevant functions, they were merely prevented from doing so by the anti-pain algorithm. However, this position can lead to odd consequences in a highly connected, algorithmically complex system, capable of reacting to diverse external inputs and internal assessments, including the influence of pseudo-random operations, i.e. exactly the kind of complex system likely to be needed for the conscious actors we are most interested in. In such highly complex systems and given a wide range of possible contexts that are unknown to the pain algorithm at the point of t_1 (perhaps not even yet defined), almost any micro-function has the possibility of being the counterfactual tipping point for diverse system-relevant functions. Indeed, one such possible future is the anti-pain algorithm being run earlier or later, which should limit the pain felt in any scenario following this second option. Taken to its natural end point, the qualia associated with all possible future functions are allowed to supervene on present micro-functions, even if only minimally. It seems we are back to something close to 'panpanpsychism' - the outcome we sought to avoid with the definition of minimal algorithms in 2.2 - and in any case, a very different position to canonical CF.

We can attempt to save the second option via 'default' counterfactuals. For instance, consider the substantial functions initiated at t_1 : the feelings associated with their partial micro-constituents might be based on the most likely or default outcome given the typical system structure. Rather than all possible counterfactual tipping points needing to be considered, only one is selected, avoiding the threat of panpanpsychism. However, the challenge lies in how to select the default counterfactual.

¹ Counterfactual accounts are a credible pathway to explore for grounding CF, given that they are one of the mainstream options available today for grounding computation implementation (contrasted with mechanistic accounts), sometimes combined with causal structure conditions or other caveats to avoid triviality critiques (Anderson & Piccinini, 2024; Sprevak, 2018; Rescorla, 2014). Pure mapping accounts are rarely applied, except in understanding the historical evolution of computation definitions. As an aside, we note that whether syntactic or semantic representation is required for computation individuation is largely a separate axis to grounding computation implementation (e.g. discussion in Fresco, 2010); the latter is the relevant target for this paper.

If the default counterfactual is a simple one, then the pain algorithm would always (or nearly always) produce pain no matter what happens afterwards, returning us back to phenomenal realism as a practical outcome, even if the underlying philosophy differs in the framing.

If the default counterfactual is selected in a more sophisticated manner, e.g. by considering the broader system structure and current state, we require a system which has sufficient visibility over the operations of its individual parts and has a rationale/threshold for excluding the more extreme counterfactual possibilities. After all, we cannot lean on an external observer to tell the system what it should be experiencing. Even if such complex processes can be incorporated into the minimum functions operating between t_1 and t_2 , implementing such system-wide visibility would be slow and computationally expensive, limiting the value of pain in driving rapid, almost automatic reactions within a system. It would also limit the types of systems that can be conscious to those which integrate such visibility at key points in all qualia algorithms.

If these concessions are granted, ‘sophisticated counterfactual calculations’ are worth exploring further but need to tread a fine line. If the counterfactuals are necessary for the experience of pain, they should be included in the minimal pain algorithm (see 2.2) and we return to earlier stages of the argument. Minimally, the explanation needs expanding to address considerations around micro-functions. If they are distinct functions, they must be complex enough to avoid trivialising towards phenomenal realism but simple enough that accelerated or parallelised anti-pain algorithms cannot prevent them from operating.

Of course, there is a solution that avoids all this discomfort. We could simply structure our CF theory such that pain is not in fact felt at t_1 , adopting for the second possible response to the thought experiment in 2.1.

2.5. Why might pain not be felt at t_1 ?

Perhaps the conclusion of a minimal pain algorithm does not lead immediately to pain being felt, but rather to ‘uncommitted pain bits’ being developed in the system. Whether those ‘pain bits’ are ultimately committed to a phenomenal ledger depends on further conditions to be specified. Pain is then felt at the point when the ‘pain bits’ are committed, not when they are developed. Within a CF perspective, it is natural for one of the necessary conditions to be some minimal functional act initiated as a result of the pain algorithm. If the experience is undone before such an act can occur, there would be no feeling felt.

On the face of it, this is a plausible account that aligns with various human experiences. For instance, we cannot perceive a state change if it occurs and reverses too quickly (e.g. in visual masking; Breitmeyer, 2008) and various phenomena have minimum thresholds for ‘just perceptible differences’ (e.g. Luce & Edwards, 1958; Green, 2023). In the human case, there is also well-documented ‘backwards memory creation’ that could cause our system at some later time period to associate the start of the pain feeling with t_1 , even though the pain did not actually begin until later. A classic example is how our brains retrospectively infer the path of a baseball, even when it is moving faster than our visual system can track (e.g. postdiction; Eagleman & Sejnowski, 2000).

In the human case, these time delays are easier to accept because conscious experiences can be read as high-level emergent phenomena and the brain has to do many computations before they are produced. Our brains are ponderous beasts. We might see ourselves stubbing a toe then have a conscious sensation of waiting, anticipating the pain, before the pain actually arrives. Time delays are also easier to accept for all the additional qualia we might scaffold around a raw pain quale, such as babies who

are unsure how much pain they ‘should’ be feeling and whether to cry until they see the reaction of their caregivers following some minor incident. Such meta-cognition plays well-known roles in our experiences of pain and would understandably be delayed (e.g. Gilam et al., 2020). However, this response to the anti-pain algorithm thought experiment requires a stronger claim: there is no initial raw pain quale, only the delayed pain quale once further system consequences, behavioural reaction, or meta-cognitive reflection have taken place. Our thought experiment also concerns a minimal pain algorithm and can be applied to an optimally simple and rapid computing system, limiting the intuitive draw of ‘delay’ analogised on the human case.

Assuming we can sustain this response, it still requires accepting the peculiar consequence that qualia can be almost free-floating, with just a tenuous link to the substantial physical computation that grounds it. The condition (or the final step in the condition) to commit the ‘pain bits’ to the ledger can be arbitrarily small and arbitrarily remote in time from the original ‘minimal’ pain algorithm that developed the pain bits in the first place in response to the original system-relevant phenomenon.

However, there is a more serious issue with this response. The pain algorithm in the thought experiment was defined as a minimal algorithm for the feeling of pain. Under this response, the true ‘minimal algorithm’ should incorporate at least one of these further conditions for committing the pain ‘bits’. The inverse algorithm would also reverse the applicable conditions. This response simply moves the problem rather than solves it. To be truly effective, we need these conditions to be such that the anti-pain algorithm becomes impossible. In other words, we move away from interpreting the thought experiment as presented and towards objecting to its validity.

Section 3. Escape routes and objections

Perhaps the anti-pain algorithm imagined in the thought experiment is not possible. We address two main routes for this objection. The first suggests that reversible computing is not possible in general for complex algorithms. The second suggests that reversible computing might be possible in some cases but not for the algorithms on which conscious experiences supervene. We briefly note another category of objections based on exotic possibilities such as block-time and reverse causality.

3.1. Reject the possibility of reversible calculations

One objection is that, once we focus on physically implemented computation, practical considerations prevent physically reversible calculations. Perfect anti-pain algorithms may not be possible, along with any other calculations above some basic level of complexity.

Engineering limits and experimental evidence as discussed, for instance, by Bormashenko (2024) show that reversible computing typically requires slowing down the computation drastically, using ultra-low temperatures, or maintaining strict isolation. Realistic computing demands (such as speed, reliability, error correction, resets, memory refresh) force heat dissipation and hence physical irreversibility for anything large-scale or general purpose. Under modest assumptions, these requirements also translate into physical limits. For instance, Earley (2022) has argued that computers necessarily dissipate entropy once they operate in an open quantum system context and thermally couple to their environment. She identifies a lower bound that corresponds to the adiabatic regime, in which the amount of entropy dissipated per computational operation is proportional to the rate of computation.

This objection can be challenged in three ways. First, arguments leaning on Landauer's principle and the second law of thermodynamics can be weakened, e.g. Bennett (2003) reminds that these principles only assign an inevitable thermodynamic cost to information destruction, rather than to information processing.² Computation can be designed to avoid over-writing, at least in modestly finite settings. Consider also how quantum computation is unitary by default: irreversibility only enters through measurement/decoherence. According to Deutsch (1985), it is the classical irreversible Turing machine that's the special restricted case; reversibility is the natural default.

Second, it is unclear that we need physical reversibility for the anti-pain algorithm to be implemented: logical reversibility plus reasonable constraints might be sufficient. The goal is to return the system to its initial state as understood by the system itself, i.e. the consciousness generated by algorithms running in the system. Any algorithm can be implemented on a reversible infrastructure, because there are Turing complete logically reversible infrastructures, such as the Toffoli and Fredkin logic gates. Although standard AND, NAND, and OR gates are irreversible (non-bijective), their logic can be implemented via reversible gates.³ By obeying Landauer's principle and expending adequate heat, an inverse function can be created that returns the logical description of the system state to the state prior to the pain algorithm being run. Provided there is nothing outside of the function that the consciousness algorithm has grip on (e.g. a measure of energy expended or temperature change over time), it would have no way of knowing the difference - and it seems oddly arbitrary to make a CF algorithm dependent on such external observations.⁴

Third, it is unclear that practical constraints on physical computation should have any bearing on computational functionalism about consciousness. In its canonical form, CF states that conscious states are identical (or at least supervene on) on the logical structure of a calculation, rather than the minutiae of how it is implemented.

Nonetheless, the messy nature of computation at the human-scale in our universe might make reversibility at the relevant scale impossible: even logical irreversibility might prove impossible given noise constraints or limitations in system isolation. Theorists could therefore sustain this objection by retreating from metaphysical necessity given the essence of consciousness (anti-pain is impossible and CF is true in all possible worlds) to nomological necessity given the practical reality of consciousness (anti-pain is impossible and CF is true given the actual laws of nature in our world). This claim places strict limits on the type of CF algorithms that can exist; their inverses must be physically impossible in our universe.

² The Landauer principle states that the erasure of one bit of information requires a minimum energy cost equal to $k_B T \ln 2$, where T is the temperature of a thermal reservoir used in the process and k_B is Boltzmann's constant. An informal statement of the second law of thermodynamics is the Clausius statement: Heat can never pass from a colder to a warmer body without some other change, connected therewith, occurring at the same time.

³ Reversibility compilers can also be constructed that allow us to make logically (and hence functionally) reversible programs on irreversible computers. For instance, as well as logic gate replacements, the compiler could replace commands like "LET X=0" (which erase information) with something reversible like "SWAP X, A" where A is a clean zero ancilla bit.

⁴ Note also that external observations introduce complexity even in the physically reversible computing context, e.g. an external clock would still show the passage of time while the pain and anti-pain algorithm ran, potentially allowing the system to infer the exact time that has passed even if unsure what happened during that 'blank' period (unless the exact amount of time is sufficient to uniquely identify a given algorithm+inverse).

3.2. Assert that consciousness algorithms are not reversible

It may simply happen to be the case (or even necessarily be the case) that conscious-generating algorithms are not reversible, even if other algorithms of comparable or larger complexity can be physically inverted. Given the points above regarding the translatability of irreversible-reversible computation, this objection requires strong motivation. It effectively requires a CF theorist to restrict their argument, such that the algorithm must be implemented on a particular substrate (e.g. using irreversible NAND gates rather than reversible Toffoli gates). How could they motivate this restriction?

One possibility is that there is some kind of grounding that happens in irreversible computations since they become interlinked with the universe. For instance, the associated decoherence might enable the ‘collapse’ of quantum superpositions and entanglement, being a necessary condition for consciousness. Perhaps a quantum embodiment variant of CF might pursue this route, but it makes CF algorithms on digital computers harder to motivate.

A second possibility is that certain features of human conscious experience require irreversible computing, such as if the phenomenal passage of time requires irreversible entropy increase. Perhaps ‘pain’ as a Platonic feeling is reversible, but any meaningful experience of temporally-extended ‘pain’ requires irreversibility. If all forms of consciousness require a sensation of time, then irreversibility becomes a necessary condition in all CF algorithms. On the other hand, the passage of time during which the pain and anti-pain algorithms ran would typically be inferrable by a sophisticated system, unless we create a highly artificial environment in which there are no observable external changes during that period of time. If the passage of time can be inferred, that might be sufficient to induce a phenomenal experience of time under CF. More worryingly, if time has a fundamental physical relevance for consciousness beyond simply keeping track of algorithmic steps, CF theorists need to develop their theory accordingly - and likely take a stance on the titanic mystery of time itself (e.g. Anderson, 2012; Emery et al., 2024).

Perhaps a form of irreversible error correction could be motivated in the same way, given that human qualia seem to have some resilience, albeit shifting this resilience from a practical benefit to a necessary condition would be an aggressive theoretical claim. Moreover, reversible forms of error correction can also be constructed (e.g. errors moved into ancilla bits in quantum computing) and it is unclear why only the irreversible forms should suffice for consciousness. Connecting and disconnecting an error correction mechanism that is never used would also seem to bring conscious experiences in/out of existence without due cause, which seems counterintuitive in line with the Olympia thought experiment and raises similar doubts about the functional relevance of consciousness (Maudlin, 1989).

Overall, this line of objection is more intriguing than convincing. Whatever theory sustains this objection, it rapidly becomes a different type of theory that loses the substrate-neutrality shine of pure CF. It will be interesting to see if any CF-friendly theorists explore this route; we are not aware of any to date.

3.3. Alternative interpretations of causality

The heart of this thought experiment draws on particular notions of causality and uncertain futures in the algorithm at t_1 . Can the scenario be defeated by rejecting those notions?

Causal history dependence

One possibility is that experiences depend on causal histories rather than solely on a system's current computational state. On this view, even if the system is returned to exactly the same internal state it occupied prior to the pain algorithm running, the rules determining consciousness still have access to the causal history of the system and can therefore distinguish whether the pain algorithm occurred. Under such a view, the anti-pain algorithm would fail to eliminate the experience because the system's past trajectory would remain relevant to the presence of pain. However, this possibility departs from the standard Turing-machine conception of computation that underlies most forms of computational functionalism.

In the classical Turing model, the evolution of the system depends only on its current state and transition rules; two systems in identical computational states must therefore evolve identically and cannot differ in virtue of *unencoded* causal histories. Allowing consciousness to depend on causal histories that are not captured in the system's current computational state would therefore weaken the claim that consciousness supervenes on the implementation of a computation in the standard sense and introduce nonlocality in time in apparent contradiction with physical laws (Maudlin, 2012).

Block-time

A second possibility is a block-time conception of the universe (Vaccaro, 2018). Under a block-time universe viewed from outside time, there is a fact of the matter about what happens at t_2 that is true at all times, including t_1 . There is no uncertainty to fuel the experiment. If the algorithm for consciousness exists (in part) outside of time, then perhaps it can draw on this information to always make the 'right' choice regarding what emotion to experience given the functional role played within a system. Or perhaps metaphysical laws simply ensure that the right qualia always supervene.

In a block-time situation, it is harder to explain why foresight does not apply in all cases. Consider the many times that our emotions are tricked by some short-term functional role when the broader system-level function should lead us down a different emotional pathway: the sensation of satiety can come too late when eating, such that it is swiftly replaced by a sensation of having over-eaten. If our consciousness is fuelled by block-time access, how can it be so incompetent?

Retrocausality

A third possibility is via reverse-causality, in which the anti-pain algorithm's completion at t_2 reaches back through time to alter experiences from t_1 . If retrocausality is possible at all in physics (Pegg, 2008), it is hard to identify in classical digital computers, placing significant limits on the type of CF that can leverage this objection.

Compound algorithms

A fourth possible objection in this vein is that the system in the thought experiment does not genuinely implement the pain algorithm at all. Instead, it implements a larger computation in which the pain algorithm is immediately followed by its inverse. On this view, the system implements a single compound algorithm whose overall functional role differs from that of the pain algorithm alone, and therefore the system never instantiates the computation associated with pain. Related concerns arise in debates about computational implementation, where theorists argue that genuine computation requires counterfactual robustness across a range of possible inputs (e.g. discussion in Schweizer, 2019). However, even if the pain algorithm is subsequently reversed, the system still executes the sequence of computational steps that constitute the pain algorithm itself. It therefore

appears natural to say that the pain computation has been implemented, followed by the implementation of a second algorithm that reverses its effects. Moreover, as the algorithms unfold, there is no certainty how they will conclude. Put simply, once the pain algorithm has run, it cannot know that it actually forms part of a larger compound algorithm that has not yet completed without introducing one of the ideas above, such as block-time or reverse-causality. If either response is correct, the thought experiment remains well-posed.

Such alternative interpretations of causality might provide creative pathways to defeating the thought experiment, but only by moving far from the counterfactual-driven, Turing machine equivalent notion of algorithms that is the commonsense substrate of consciousness in CF.

Conclusion

Under conventional interpretations of computation, it is possible to design inverse functions that return a system exactly to the state it was in prior to some minimal pain algorithm. If conscious experiences are identical to (or at least supervene on) the appropriate algorithms in line with computational functionalism (CF), it is reasonable to ask what happens to those experiences where a ‘pain’ algorithm is followed immediately by an inverse ‘anti-pain’ algorithm. This paper shows how exploring this scenario pressures at least one concession on CF (summarised in Table 1).

If pain is felt before the anti-pain algorithm concludes, theorists are pressured either to permit micro-functions such that CF approximates phenomenal realism (with pain experienced as an absolute phenomenon regardless of any functional relevance) or to restrict consciousness only to systems that have system-wide visibility mechanisms to forecast likely functional relevance. On the other hand, if pain is argued not to be felt, theories risk permitting free-floating qualia, with only a tenuous link to the spatiotemporal physicality of its supervenience base. Multiple responses are viable, leading to different directions for future theory development.

The thought experiment can also be objected to by rejecting the possibility of such inverse algorithms. However, these objections typically lead to weakening the notion of computation in CF, such as reducing its claims to metaphysical necessity or its substrate-neutrality. Given these different avenues that CF theorists might pursue, we hope that this anti-pain thought experiment helps clarify the type of CF that different theorists support.

We would also suggest non-CF theorists explore equivalent versions of this thought experiment. Parallels with ‘forgotten’ pain in the study of ethics and medicine may be similar enough to anti-pain that exact algorithmic inversion is not necessary to challenge other theories of consciousness, particularly other functionalist theories (e.g. Lee et al., 2020; Pryor et al., 2025). Perhaps different temporal supervenience bases for qualia under specific theories restructure the argument enough to make certain pathways trivial to accept, but this is a possibility that requires proper exploration.

Other extensions include how the scenario might vary if the anti-pain algorithm being run is dependent on an uncertain external factor, such as a cosmic ray or a person deciding to press a button. The relationship between clock time, system time, and phenomenal time also merits further exploration under a CF perspective. What if the algorithm is delayed arbitrarily, but during a period of time when no irreversible external action occurs, so the pain and perhaps its internal functional consequences can still be inverted by an anti-pain algorithm?

Table 1: Landscape of responses to the anti-pain algorithm thought experiment (summary)

From a functionalist perspective*, is pain felt after a minimal pain algorithm concludes and before an anti-pain algorithm concludes?		
Answer	Options to sustain the response	Pressures on theorists from that option
1. Yes	a. Embrace 'thin' functions as grounding pain (e.g. micro-functions = micro-qualia)	Arbitrarily thin functions become viable for qualia, challenging intuitions around material causal relevance within a system architecture and pushing the theory towards a form of phenomenal realism.
	b. Allow generous counterfactuals given what system-relevant impact 'thin' functions might lead to in the future	In the limit in complex systems, any 'thin' function could lead to many outcomes and diverse qualia. All supervene on the 'thin' function ('panpanpsychism').
	c. Impose a blunt default counterfactual that gives 'thin' functions relevance	You always get pain regardless of what actually happens ('phenomenal realism').
	d. Compute a sophisticated counterfactual to give 'thin' functions relevance	Requires system-wide visibility to estimate what is likely to happen, both computationally expensive and restrictive on feasibly conscious systems.
	e. Reject need for functions to have relevance in a system architecture	If consequences don't matter, we depart classical functionalism towards phenomenal realism.
2. No	a. Pain is felt when pain bits are 'committed' following a functional impact, not just when pain bits are 'prepared' by a pain algorithm	Qualia can become almost free-floating, as 'committing' can happen arbitrarily far from the original pain algorithm.
	b. Genuine pain requires meta-cognitive assessment of the pain percepts before it is felt, during which time system-relevant functions can take place	Raw pain does not exist in contradiction to apparent human phenomenology; only meta-cognised pain exists.
	<i>Moreover, in both 'No' responses, the minimum pain algorithm should in fact be redefined to include the committing/meta-cognition processes (see 2.2 for details), risking a return to a 'Yes' position</i>	
3. Reject anti-pain	a. Reversible calculations of any scale are impossible in our universe, e.g. due to entropy and environment interactions	Places strong restrictions on what counts as computation. Departs from canonical functionalism in which logical states/transitions matter, not substrate-specific aspects of physical implementation.
	b. Conscious algorithms require irreversible constructions, e.g. given phenomenology of time	Substrate-neutrality is weakened, e.g. an algorithm is conscious under irreversible NAND gates but not under reversible Toffoli gates. Physicalist concepts of time may need introducing into the theory.
	c. Causality prevents the described anti-pain, e.g. via block-time, reverse-causality, causal history dependence (time-nonlocality)	All these notions depart from the Turing Machine compatible notion of computation, requiring novel theoretical motivation.

*Computational functionalism grounds phenomenology in virtue of the functional role an algorithm plays in its system. One alternative is phenomenal realism whereby a particular physical or informational object inherently conveys qualia in the present state, regardless of dynamic mechanisms that are defined across previous/subsequent states (evolved systems or dynamic functions might still exploit such qualia but would not constitute them).

An alternative thought experiment could retain a memory trace that the pain algorithm occurred (e.g. a single bit indicator against a time-stamp), even though the more complex algorithm associated with the feeling itself is itself reversed. Since the system knows that pain occurred in that setting (and would have remained present without the anti-pain algorithm), it might choose to use that information to shape future decisions. But if so, what are the functional and phenomenal differences, if any, between the single-bit indicator and the full pain algorithm? What if the single-bit indicator were only 'discovered' later after an arbitrary delay? This variant could probe intuitions disconnecting valence from instrumental goals and the temporal supervenience basis of qualia.

We close by speculating on what 'anti-pain qualia' might feel like if experienced on its own, without requiring its annihilation upon contact with a previous 'pain qualia'. Unless someone can identify an

inverse function of another type that humans have a track record of experiencing, this seems to be a challenging speculation.

Would anti-pain be like rewind cinema, where the experience of pain somehow ‘plays backwards’? Would it be a pleasurable experience of exact magnitude and quality that it counterbalances the pain qualia, based on metrics defined in a continuous symmetrical space of valence? Or perhaps it’s a dreamless blank, as might occur if the architectures necessary for consciousness only work forwards, e.g. the right kinds of loops or global integration do not occur backwards?

The differences matter under certain positions in ethics. For instance, under a naïve utilitarian summation of valence across all conscious moments, the original pain might be fully counterbalanced if anti-pain feels symmetrically pleasurable, but not if it feels like a dreamless blank (depending on how the timing of qualia experiences is resolved in our thought experiment). In a dreamless blank, future decisions would still be unaffected by the presence of a past - but now perfectly reversed and forgotten - pain, but that moment of pain might still exist, permanently captured in the universal utilitarian ledger.

The most likely feeling perhaps is some ‘alien qualia’ - unimaginable to a human brain that has never experienced an inverse function. Perhaps hard-to-imagine phenomenology is simply table-stakes in what Schwitzgebel (2014) has described as the ‘crazyist’ metaphysics of mind, operating in a universe that likely permits a vast ‘terra incognita’ of peculiar conscious experiences (Shanahan, 2025). But we can still have fun trying.

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